3.1 Constrained Optimization

Lagrange multipliers, the primal, and the dual.

1. Extra conditions on $\boldsymbol{\theta}$

Unconstrained: $\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$. Constrained: the same min, but $\boldsymbol{\theta}$ must sit in

$$S=\{\boldsymbol{\theta}: g_i(\boldsymbol{\theta})=0\text{ or }g_i(\boldsymbol{\theta})\le 0\}.$$

A box $-1\le\theta_j\le 1$ is already a constraint set. The unconstrained min can fall **outside** the box. The constrained min is then on the boundary (the star, not the circle, on the usual contour cartoon).

Lab 3 is one equality: $4\theta_1+2\theta_2=12$, quadratic J. You will write a Lagrangian and set every partial to zero.

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2. The Lagrangian

Turn the constrained problem into an unconstrained one in more variables. For inequalities $g_i(\boldsymbol{\theta})\le 0$,

$$\mathcal{L}(\boldsymbol{\theta},\boldsymbol{\lambda}) = J(\boldsymbol{\theta})+\boldsymbol{\lambda}^\top\mathbf{g}(\boldsymbol{\theta})$$

(sign conventions differ; stay consistent). $\boldsymbol{\lambda}$ are **multipliers**. They are the prices of the constraints. If a constraint is slack, its multiplier is zero (KKT complementary slackness). If it is tight, the multiplier says how much J would improve if you relaxed it.

Why convert? Most algorithms, and most theory, are written for unconstrained maps. The Lagrangian is the conversion.

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3. Primal and dual

The **primal** is the original problem in $\boldsymbol{\theta}$. The **dual** is an optimization in $\boldsymbol{\lambda}$, obtained by minimizing $\mathcal{L}$ in $\boldsymbol{\theta}$ first (when that min is easy). Weak duality: dual objective $\le$ primal objective. Strong duality (typical for convex problems that satisfy a constraint qualification): the values match, and the multipliers at the saddle recover the primal $\boldsymbol{\theta}$.

Week 3 notes **3.2** and **3.3** do this for an LP and a QP. The SVM dual in Module 3 is the same move with a quadratic J.